

40.520 Stochastic Models

Brownian Motion

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Outline

1. Definition of Brownian Motion
2. Hitting Times and Maximum Variable
3. Moment Generating Function
4. Drifted Brownian Motion

Definition of Brownian Motion

Symmetric Random Walk

- The symmetric random walk can be represented by

$$Z_n = \sum_{i=1}^n X_i$$

with $\{X_i, i \geq 1\}$ being a sequence of independent random variables, whose distribution is given by

$$P(X_i = 1) = P(X_i = -1) = 1/2$$

- The symmetric random walk is not only a **Markov process** but also a **Martingale process**.

Independent and Stationary Increments

Take a sequence of integers:

$$0 = i_0 < i_1 < \dots < i_n.$$

- Independent increments:

$$Z_{i_1} - Z_{i_0}, Z_{i_2} - Z_{i_1}, \dots, Z_{i_n} - Z_{i_{n-1}}$$

Any two disjoint increments

are independent.

- Stationary increments: the distribution of

$$Z_{n+m} - Z_n$$

does not depend on n .

Scaled Random Walk

- Rescale time and space for the random walk model. That is, let

$$Y_t = \Delta x \cdot \sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i.$$

- **Mean:**

$$E[Y_t] = 0$$

- **Variance:**

$$\text{Var}[Y_t] = (\Delta x)^2 \lfloor \frac{t}{\Delta t} \rfloor$$

$$Z_t = \sum_{i=1}^t X_i, \quad X_i = \begin{cases} 1 & \text{w.p. } \frac{1}{2} \\ -1 & \text{w.p. } \frac{1}{2} \end{cases}$$

$$\textcircled{1} \text{ (scaled step size) } \Delta X_i = \begin{cases} -\Delta x & \text{w.p. } \frac{1}{2} \\ \Delta x & \text{w.p. } \frac{1}{2} \end{cases}$$

$$\textcircled{2} \text{ (scaled time step) } \left[\frac{1}{\Delta t} \right] \quad \left[\frac{t}{\Delta t} \right]$$

$$Y_t = \sum_{i=1}^{\lfloor t/\Delta t \rfloor} \Delta X \cdot X_i$$

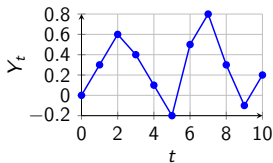
$$\therefore E[Y_t] = E\left[\Delta X \sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i\right] = \Delta X \sum_{i=1}^{\lfloor t/\Delta t \rfloor} E[X_i] = 0. \quad \#$$

$$\begin{aligned} \therefore \text{Var}[Y_t] &= \text{Var}\left[\Delta X \sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i\right] = (\Delta X)^2 \text{Var}\left[\sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i\right] \\ &= (\Delta X)^2 \sum_{i=1}^{\lfloor t/\Delta t \rfloor} \text{Var}[X_i] = (\Delta X)^2 \left[\frac{t}{\Delta t} \right] \text{Var}[X_i] \\ &= (\Delta X)^2 \left[\frac{t}{\Delta t} \right] \quad \# \end{aligned}$$

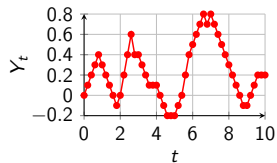
Limits of Scaled Random Walk

- We now let Δx and Δt go to 0 in such a way that

$$\Delta x = \sqrt{\Delta t}.$$



Unscaled Random Walk ($\Delta t = 1$)



Scaled Random Walk ($\Delta t = 0.2$)

Limits of Scaled Random Walk

- By the central limit theorem,

$$Y_t \sim N(0, \sqrt{t}) \text{ as } \Delta x \rightarrow 0.$$

$$\begin{aligned} Y_t &= \Delta x \sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i \\ &= \sqrt{\Delta t} \sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i \\ &= \sqrt{\Delta t} \cdot \sqrt{\frac{t}{\Delta t}} \frac{\sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i}{\sqrt{\frac{t}{\Delta t}}} = \sqrt{t} \frac{\sum_{i=1}^{\lfloor t/\Delta t \rfloor} X_i}{\sqrt{\frac{t}{\Delta t}}} \end{aligned}$$

By CLT $\sim N(0, 1)$

Brownian Motion

A stochastic process $\{X_t, t \geq 0\}$ is said to be a (standard) Brownian motion process if:

- $X_0 = 0$;
- It has stationary and independent increments;
- For every $t > 0$, X_t is normally distributed with mean 0 and variance t ;
- It has continuous sample paths.

Brownian Motion as Markov and Martingale Processes

- The independent increment assumption implies that Brownian motion is a Markov process:

$$P\{X_{t+s} \leq a \mid X_u, 0 \leq u \leq s\} = \Phi\left[\frac{a - X_s}{\sqrt{t}}\right],$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution.

- It also implies that Brownian motion is a martingale (continuous-time):

$$E[X_{t+s} \mid X_u, 0 \leq u \leq s] = X_s.$$

$$\begin{aligned}
P(X_{t+s} \leq a \mid X_u, 0 \leq u \leq s) &= P(X_{t+s} - X_s \leq a - X_s \mid X_u, 0 \leq u \leq s) \\
&= P(X_t \leq a - X_s \mid X_u, 0 \leq u \leq s) \\
&= P\left(\frac{X_t}{\sqrt{t}} \leq \frac{a - X_s}{\sqrt{t}}\right) \\
&= \Phi\left(\frac{a - X_s}{\sqrt{t}}\right) \neq
\end{aligned}$$

$$\begin{aligned}
E[X_{t+s} \mid X_u, 0 \leq u \leq s] &= E[X_{t+s} - X_s + X_s \mid X_u, 0 \leq u \leq s] \\
&= E[X_{t+s} - X_s \mid X_u, 0 \leq u \leq s] + E[X_s \mid X_u, 0 \leq u \leq s] \\
&= E[X_t \mid X_u, 0 \leq u \leq s] + X_s \\
&= 0 + X_s \\
&= X_s \neq
\end{aligned}$$

Covariance of Brownian Motion

Covariance: if $s < t$,

$$\text{Cov}[X_s, X_t] = \text{Var}(X_s) = s.$$

$$\begin{aligned}\text{Cov}(X_s, X_t) &= \text{Cov}(X_t - X_s + X_s, X_s) \\ &= \text{Cov}(X_t - X_s, X_s) + \text{Cov}(X_s, X_s) \\ &= 0 + \text{Var}(X_s) \\ &= s. \quad \# \end{aligned}$$

Example

- For $t_1 < t_2 < t_3$, compute

$$E[X_{t_1} \cdot X_{t_2} \cdot X_{t_3}].$$

$$\begin{aligned}
E[X_{t_1} \cdot X_{t_2} \cdot X_{t_3}] &= E[X_{t_1} \cdot X_{t_2} \cdot (X_{t_3} - X_{t_2} + X_{t_2})] \\
&= E[X_{t_1} \cdot X_{t_2} \cdot (X_{t_3} - X_{t_2})] + E[X_{t_1} \cdot X_{t_2}^2] \\
&= E[X_{t_1} \cdot X_{t_2}] \underbrace{E[X_{t_3} - X_{t_2}]}_{=0} + E[X_{t_1} \cdot X_{t_2}^2] \\
&= E[X_{t_1} \cdot (X_{t_2} - X_{t_1} + X_{t_1})^2] \\
&= E[X_{t_1} \cdot [(X_{t_2} - X_{t_1})^2 + 2X_{t_1}(X_{t_2} - X_{t_1}) + X_{t_1}^2]] \\
&= \underbrace{E[X_{t_1} \cdot (X_{t_2} - X_{t_1})^2]}_{=0} + 2 \underbrace{E[X_{t_1}^2 (X_{t_2} - X_{t_1})]}_{=0} + E[X_{t_1}^3] \\
&= E[X_{t_1}^3] \neq
\end{aligned}$$

Hitting Times and Maximum Variable

Distribution of Hitting Time

Let T_a be the first time the Brownian motion process hits a level $a > 0$. Then

$$P\{T_a \leq t\} = 2P\{X_t \geq a\} = \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^{+\infty} e^{-\frac{y^2}{2}} dy.$$

which implies that the probability density function of T_a is

$$f_{T_a}(t) = \frac{a}{\sqrt{2\pi}} t^{-\frac{3}{2}} e^{-\frac{a^2}{2t}}.$$

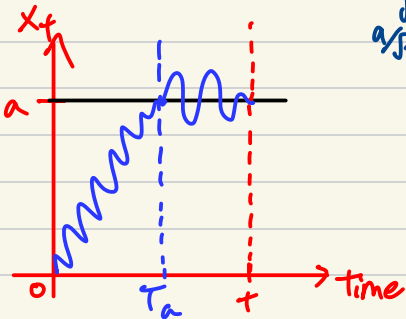
$$P(X_t \geq a) = \underbrace{P(X_t \geq a | T_a \leq t)}_{\frac{1}{2}} \cdot P(T_a \leq t) + \underbrace{P(X_t \geq a | T_a > t)}_0 \cdot P(T_a > t)$$

$$P(T_a \leq t) = 2P(X_t \geq a)$$

$$= 2P\left(\frac{X_t}{\sqrt{t}} \geq \frac{a}{\sqrt{t}}\right)$$

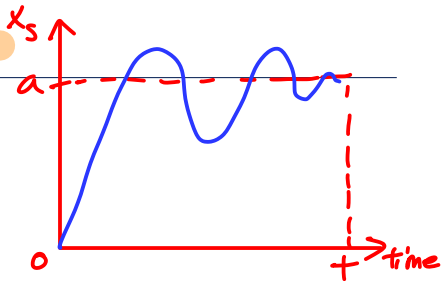
$$= 2 \int_{\frac{a}{\sqrt{t}}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy = F_{T_a}(t)$$

$$\therefore f_{T_a}(t) = \frac{d}{dt} F_{T_a}(t) \neq$$



Maximum Value of a Brownian Motion

$$P\left(\max_{0 \leq s \leq t} X_s \geq a\right) = P(T_a \leq t)$$



- For $a > 0$,

$$P\left\{\max_{0 \leq s \leq t} X_s \geq a\right\} = P\{T_a \leq t\} = \frac{2}{\sqrt{2\pi}} \int_{a/\sqrt{t}}^{+\infty} e^{-\frac{y^2}{2}} dy.$$

Example: Hitting Probability of Brownian Motion

- Note that Brownian motion is a martingale process. We still can use the martingale stopping theorem to compute

$$P(X_T = A) = \frac{B}{A+B}$$

where T is the first time the Brownian motion X hits A or $-B$.

$$\begin{aligned} E[X_T] &= P(A) \cdot A + P(-B) \cdot (-B) = 0 \\ &= P(A) \cdot A - [1 - P(A)]B = 0 \end{aligned}$$

$$\therefore P(A) = \frac{B}{A+B}$$

Moment Generating Function

Moment Generating Function (MGF)

Definition: For a random variable X , the moment generating function

$$M_X(t) = \mathbb{E} \left[e^{tX} \right],$$

where t is a real number.

$$E[e^{tX}] = \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx$$

MGF: Examples

Each distribution has a unique MGF. For example:

Distribution	PDF / PMF	MGF $M_X(t)$
Normal $N(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$e^{\mu t + \frac{1}{2}\sigma^2 t^2}$
Exponential (λ)	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{\lambda}{\lambda - t}, t < \lambda$
Poisson (λ)	$e^{-\lambda} \frac{\lambda^k}{k!}$	$e^{\lambda(e^t - 1)}$
Binomial (n, p)	$\binom{n}{k} p^k (1-p)^{n-k}$	$(1 - p + pe^t)^n$

$$X \sim N(0, 1)$$

$$M_X(t) = E[e^{tX}]$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tx - \frac{x^2}{2}} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 - 2tx + t^2) + \frac{t^2}{2}} dx$$

$$= \frac{e^{t^2/2}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2} dx$$

$$= \frac{e^{t^2/2}}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = e^{t^2/2} \quad \#$$

Key Properties

1. Sum of Independent RVs:

$$E[e^{t(X+Y)}] = E[e^{tX} \cdot e^{tY}] = E[e^{tX}] \cdot E[e^{tY}]$$

If $Z = X + Y$ with $X \perp Y$, then $M_Z(t) = M_X(t) \cdot M_Y(t)$

2. Linear Transformation:

$$M_{aX+b}(t) = e^{bt} M_X(at)$$

3. Uniqueness:

If $M_X(t) = M_Y(t)$ for all t , then X and Y have the same distribution.

4. Moment Extraction:

$$\mathbb{E}[X^n] = \left. \frac{d^n}{dt^n} M_X(t) \right|_{t=0}$$

$$\left. \frac{d E[e^{tX}]}{dt} \right|_{t=0} = E[X e^{tX}]|_{t=0} = E[X]$$

Example: Sum of Independent Normals

Let $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$, independent. Show

$$Z = X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

↓

$$\therefore M_Z = M_X \cdot M_Y$$

$$= (e^{\mu_1 t + \frac{1}{2} \sigma_1^2 t^2}) (e^{\mu_2 t + \frac{1}{2} \sigma_2^2 t^2})$$

$$= e^{(\mu_1 + \mu_2)t + \frac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2} \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

≠

Brownian Motion with Drift

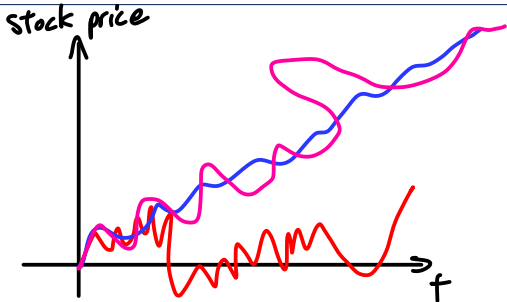
Brownian Motion with Drift

Standard Brownian Motion B_t :

- No trend: $\mathbb{E}[B_t] = 0$
- Variance grows linearly: $\text{Var}(B_t) = t$

But real-world processes have:

Process = Deterministic Trend + Random Noise



Brownian Motion with Drift

- Let $\{B_t, t \geq 0\}$ be a standard Brownian motion process. If we attach it with a deterministic drift, i.e., let

$$X_t = \sigma B_t + \mu t,$$

we say that $\{X_t, t \geq 0\}$ is a Brownian motion with drift μ and volatility σ .

- Mean:** $E[X_t] = \mu t$

- Variance:** $Var[X_t] = \sigma^2 t$

$$\begin{aligned} E[X_t] &= E[\sigma B_t + \mu t] \\ &= \sigma E[B_t] + \mu t \\ &= \mu t \quad \neq \end{aligned}$$

$$\begin{aligned} Var[X_t] &= Var[\sigma B_t] \\ &= \sigma^2 Var[B_t] = \sigma^2 t \quad \neq \end{aligned}$$

Drifted Brownian Motion

The above drifted Brownian motion has the following properties:

- $X_0 = 0$;
- It has stationary and independent increments;
- X_t is normally distributed with mean μt and variance $\sigma^2 t$.

Geometric Brownian Motion

If $\{X_t, t \geq 0\}$ is a Brownian motion process with drift μ and volatility σ , then the process $\{Y_t, t \geq 0\}$ defined by

$$Y_t = e^{X_t}$$

is called a geometric Brownian motion.

- Positivity
- Proportional random changes (returns)
- Matches real data (lognormal distribution)

Example

Let $Y_t = e^{X_t}$, where $\{X_t, t \geq 0\}$ is a Brownian motion process with drift μ and volatility σ . Show

$$E[Y_t | Y_s = y] = y \cdot e^{(t-s)(\mu + \sigma^2/2)}$$

$$X_t - X_s \sim N(\mu(t-s), \sigma^2(t-s))$$

$$\begin{aligned} E[Y_t | Y_s = y] &= E[e^{X_t} | Y_s = y] \\ &= E[e^{X_t - X_s + X_s} | Y_s = y] \\ &= E[e^{X_t - X_s} | Y_s = y] \cdot E[e^{X_s} | Y_s = y] \\ &= E[e^{X_t - X_s}] \cdot y \\ &= y(e^{(t-s)(\mu + \sigma^2/2)}). \quad \# \end{aligned}$$